Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	LTVIP2026TMIDS80513
Project Name	TransLingua: AI-Powered Multi-Language Translator
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Form • Registration through Gmail • Registration through LinkedIn
FR-2	User Confirmation	• Confirmation via Email • Confirmation via OTP
FR-3	**Core AI Translation**	• Text-to-Text Translation using Gemini API • Source and Target Language Selection • Context-Aware Tone Toggles (Professional/Academic/Casual)
FR-4	**Document & Scenario Handling**	• File Upload for Academic Paper Translation • Preservation of LaTeX and Citations • Business Brand Glossary Integration
FR-5	**Mobile Optimization (Travel)**	• High-Contrast UI for mobile travelers • Voice-to-Text Input Translation • Offline Phrasebook for common travel needs

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Streamlit interface must be intuitive enough for a traveler to translate a sign in under 3 clicks.
NFR-2	**Security**	Secure handling of API keys and encryption of user-uploaded academic or business documents.
NFR-3	**Reliability**	The system must maintain high translation accuracy (verified by back-translation checks) to ensure trust in professional settings.
NFR-4	**Performance**	Translation requests to the Gemini API should return results within 2–5 seconds to support real-time travel needs.

NFR-5	**Availability**	The web application should be accessible 99% of the time, especially for researchers working across different time zones.
NFR-6	**Scalability**	The architecture must support an increasing number of concurrent users during peak travel seasons or academic deadlines.